THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A student models a falling ball ignoring air resistance. For which object is the model least accurate? 1

(a) a stone **(b)** an iron ball **(c)** a steel bolt **(d)** a feather

2. A solution is prepared by dissolving 30 g of sugar in 120 g of water. The mass percentage is: 1

(a) 12.5 per cent **(b)** 30 per cent **(c)** 20 per cent **(d)** 25 per cent

3. In an experiment 5.6 g of CaO reacts with 4.4 g of CO2 to form CaCO3. The mass of CaCO3 formed is: 1

(a) 9.6 g **(b)** 11 g **(c)** 1.2 g **(d)** 10 g

4. An atom has the configuration 2, 8, 6. The number of protons is: 1

(a) 16 **(b)** 8 **(c)** 14 **(d)** 6

5. A pendulum model assumes no friction. The actual pendulum: 1

(a) swings forever **(b)** stops at once
(c) slowly stops swinging **(d)** speeds up

6. A mixture of two liquids X (b.p. 78 degrees Celsius) and Y (b.p. 118 degrees Celsius) is best separated by: 1

(a) fractional distillation **(b)** filtration
(c) sublimation **(d)** crystallisation

7. The formula of aluminium sulphate is: 1

(a) $AlSO_4$ **(b)** $Al(SO_4)_3$ **(c)** $Al_3(SO_4)_2$ **(d)** $Al_2(SO_4)_3$

8. An element has 3 shells and 2 valence electrons. Its atomic number is: 1

(a) 12 **(b)** 8 **(c)** 10 **(d)** 14

9. The density of an object is 2.5 g per cubic cm. In kg per cubic metre this is: 1

(a) 2500 **(b)** 250 **(c)** 25 **(d)** 25000

10. A student has a mixture of sand, salt and camphor. The correct sequence of separation is: 1

(a) evaporation, sublimation, filtration
(b) dissolving, sublimation, evaporation
(c) filtration, sublimation, evaporation
(d) sublimation, dissolving and filtering, evaporation

11. Which of these pairs forms an ionic compound? 1

(a) N and H **(b)** C and O **(c)** Mg and Cl **(d)** C and H

12. Chlorine has isotopes of mass 35 u (75 per cent) and 37 u (25 per cent). The average atomic mass is: 1

(a) 35 u **(b)** 36 u **(c)** 35.5 u **(d)** 37 u

13. A runner covers 100 m in 12.5 s. The estimated speed is: 1

(a) 8 m/s **(b)** 1.25 m/s **(c)** 80 m/s **(d)** 12.5 m/s

14. How many grams of salt are present in 250 g of a 4 per cent salt solution? 1

(a) 10 g **(b)** 4 g **(c)** 40 g **(d)** 25 g

15. Ammonia contains 82.35 per cent nitrogen. In 34 g of ammonia the mass of nitrogen is: 1

(a) 6 g **(b)** 28 g **(c)** 17 g **(d)** 14 g

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A model assumes frictionless motion. Give one situation where it is acceptable and one where it fails. 2

17. Calculate the mass of sugar required to prepare 500 g of 8 per cent solution. 2

18. In an experiment 2.0 g of hydrogen reacts with 16 g of oxygen to form water. Find the mass of water and verify the law of conservation of mass. 2

19. An atom has 12 protons and 12 neutrons. Find its mass number and electronic configuration. 2

20. Why do scientists express large numbers in scientific notation? Write 300000000 in this form. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A student measures the time period of a pendulum by timing 20 oscillations as 32 s. Find the time period and say why 20 oscillations are timed. 3

22. A 200 g solution has 36 g salt. Find the percentage. If 100 g water is evaporated find the new percentage. 3

23. A compound of carbon and oxygen has C : O = 3 : 8 by mass. Find the amounts of C and O in 55 g of the compound. 3

24. An element has 20 protons and 20 neutrons. Find its mass number, electronic configuration, valency and name. 3

OR

Explain the importance of scientific notation, using examples of the size of an atom and the distance to the Sun. 3

25. A mixture of NaCl and KNO3 is dissolved in hot water and cooled. Which crystallises first and why? 3

OR

State three differences between sodium chloride and carbon tetrachloride. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A pendulum experiment gave: length (cm) 25, 50, 75, 100 and time for 10 oscillations (s) 10, 14.2, 17.4, 20. Calculate the time periods, comment on the relationship and mention the assumptions. 5

OR

A student has three solutions A, B, C of the same solute with masses of solute 5 g in 95 g water, 10 g in 90 g water and 15 g in 85 g water. Find the percentage of each, rank them and say what happens on mixing A and C (equal masses). 5

27. Verify the law of constant proportions using water: 9 g of water gives 1 g hydrogen and 8 g oxygen, and 18 g of water gives 2 g hydrogen and 16 g oxygen. State the law. 5

OR

An atom X has atomic number 13 and mass number 27. Find the electrons, protons, neutrons, configuration, valency, the ion it forms and the formula of its oxide. 5

END OF QUESTION PAPER — Total: 50 Marks

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